

01^{/18}

TAR REIN THWAY
PIUS 20220022
25TH MAY 2026

FOOD CLASSIFICATION

F I N A L
P R O J E C T

ALL PHOTOS ARE FROM THE DATASET

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A U D I E N C E

PROFESSOR & CLASSMATES WHO ARE ALSO LEARNING CNN.

CONTENT

1. Project Overview
2. Dataset
3. Preprocessing Step
4. Models
5. Evaluation
6. UI Demonstration



PROJECT OVERVIEW

CNN - Transfer Learning - Multi Class Image Classification

One click analysis - Challenging Enough to Promote Learning - Real World Usability

Limitations

Technical Details

Conceptual Depth

Remedy

GitHub + Webapp Link in Chat

Presenting



DATASET

Title: Food 101 - Mining Discriminative Components with Random Forest



- Published - 2014
- 101,000 Images
- 101 Class Labels
- 75% Training + 25% Test
- 512 Pixel Images
- High Intra-Class Variations
- Inter-class Similarities
- Uncleaned Raw Images
- Similarity to ImageNet Dataset

INTRA-CLASS VARIATIONS

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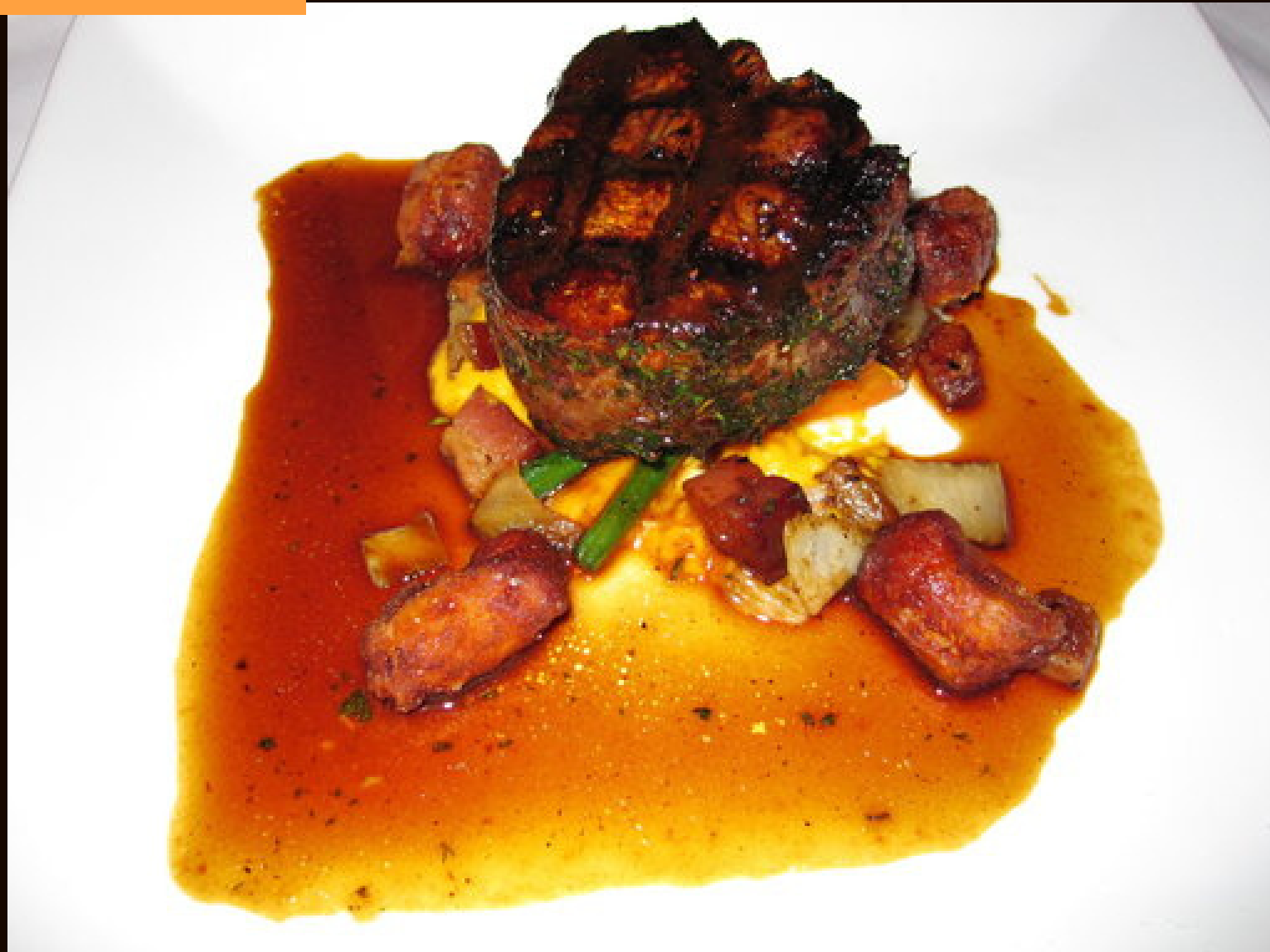
Apple Pie



INTER-CLASS SIMILARITY

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Filet Mignon



Steak



PRE-PROCESS

- Download Dataset
- Splitting Dataset
- Read/Decode/Resize
- Data Augmentation
- Preprocess (Model based)
- Batching



PRE-PROCESS

Basic Configurations

Image Size - 224

Batch Size - 64

Splitting Dataset

Train : 75% (750 images/class)

Test: 12.5% (125 images/class)

Validation: 12.5% (125 images/class)

Data Augmentation

Horizontal Flips

Cropping

Color Jittering



Data Augmentation

Original (Top) vs Augmented Image (Bottom)





MODEL ARCHITECTURE

- Input Layer
- Base Model
- Global Average Pooling
- Batch Normalization
- Dropout
- Dense (101, 'softmax')

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DENSENET201

Base Model



Code

`tf.keras.applications.DenseNet201`

700 layers with 5 major blocks.

18.3 Million trainable parameters

77.3% accuracy on ImageNet



TRAINING STAGES

Classification Head

Freeze the base model

Epoch 5

Adam (LR=2e-3)

Fine Tuning

Progressive Unfreezing

30 Epochs (10 for each phase)

AdamW (custom learning rate + weight decay = 1e-4)

Phase 1 - Highest Abstraction Block

Phase 2 - Mid Level Blocks

Phase 3 - Full Model

LEARNING RATE SCHEDULE

LR_START = $1e-5$

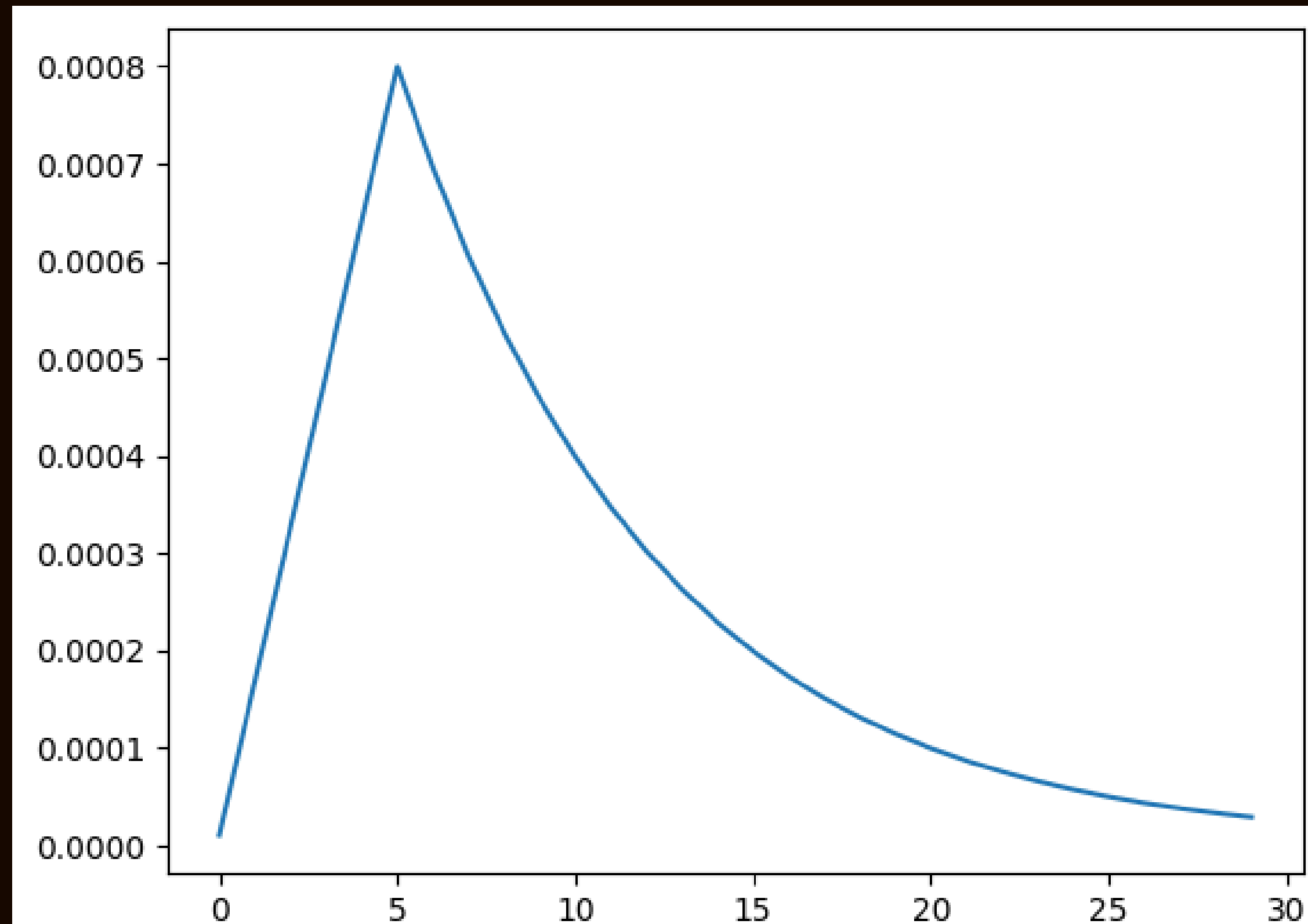
LR_MAX = $8e-4$

LR_MIN = $1e-7$

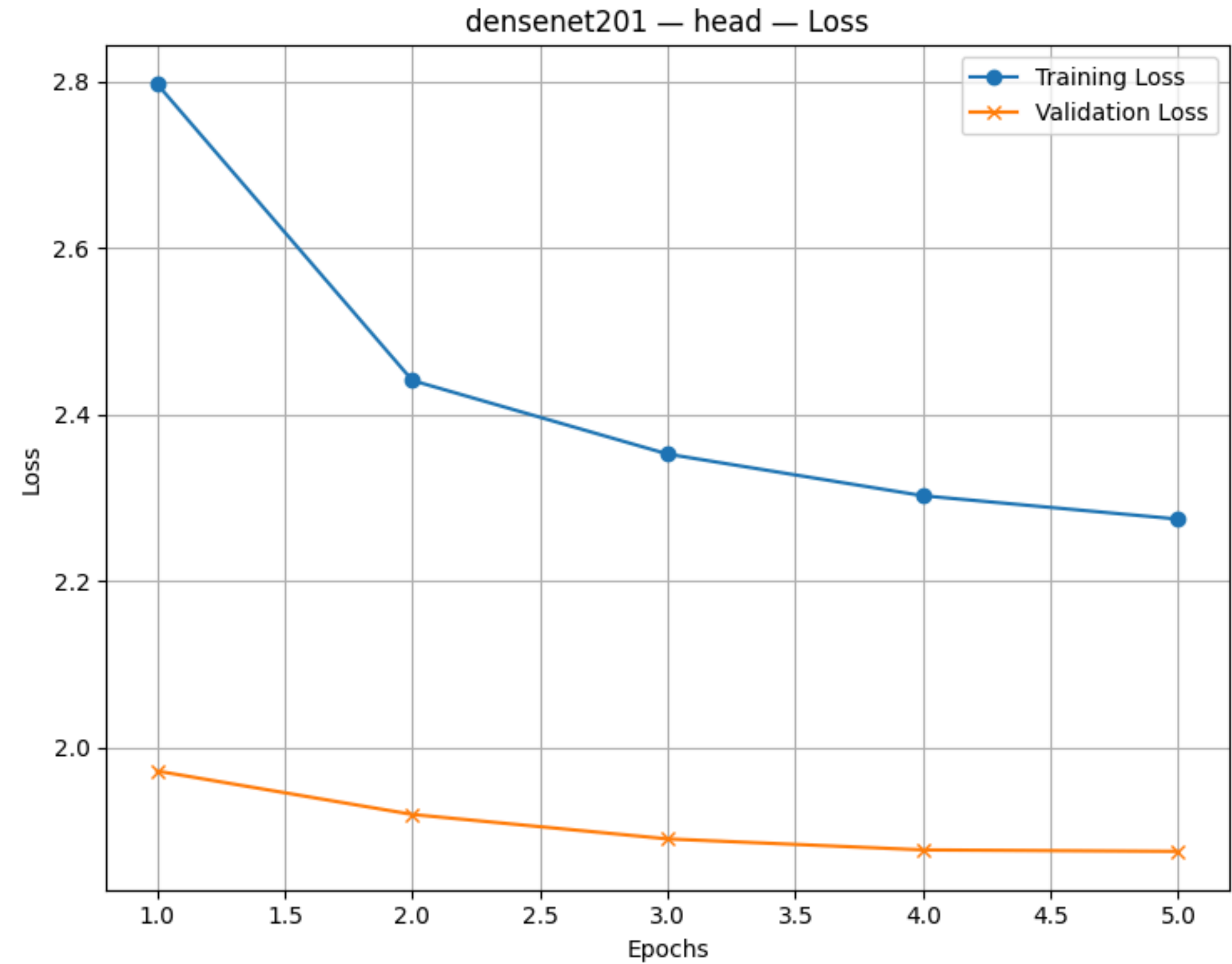
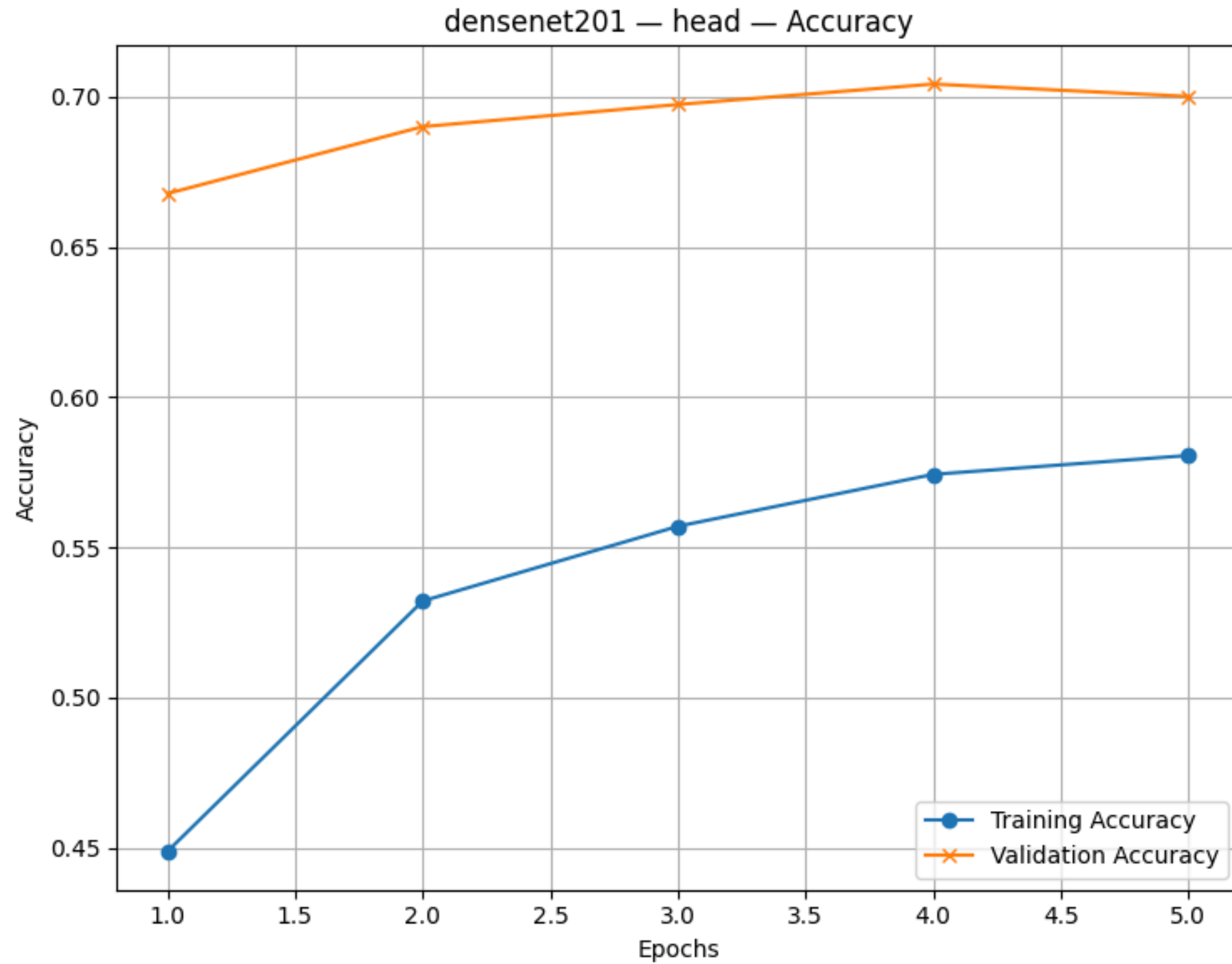
LR_RAMPUP_EPOCHS = 5

LR_SUSTAIN_EPOCHS = 0

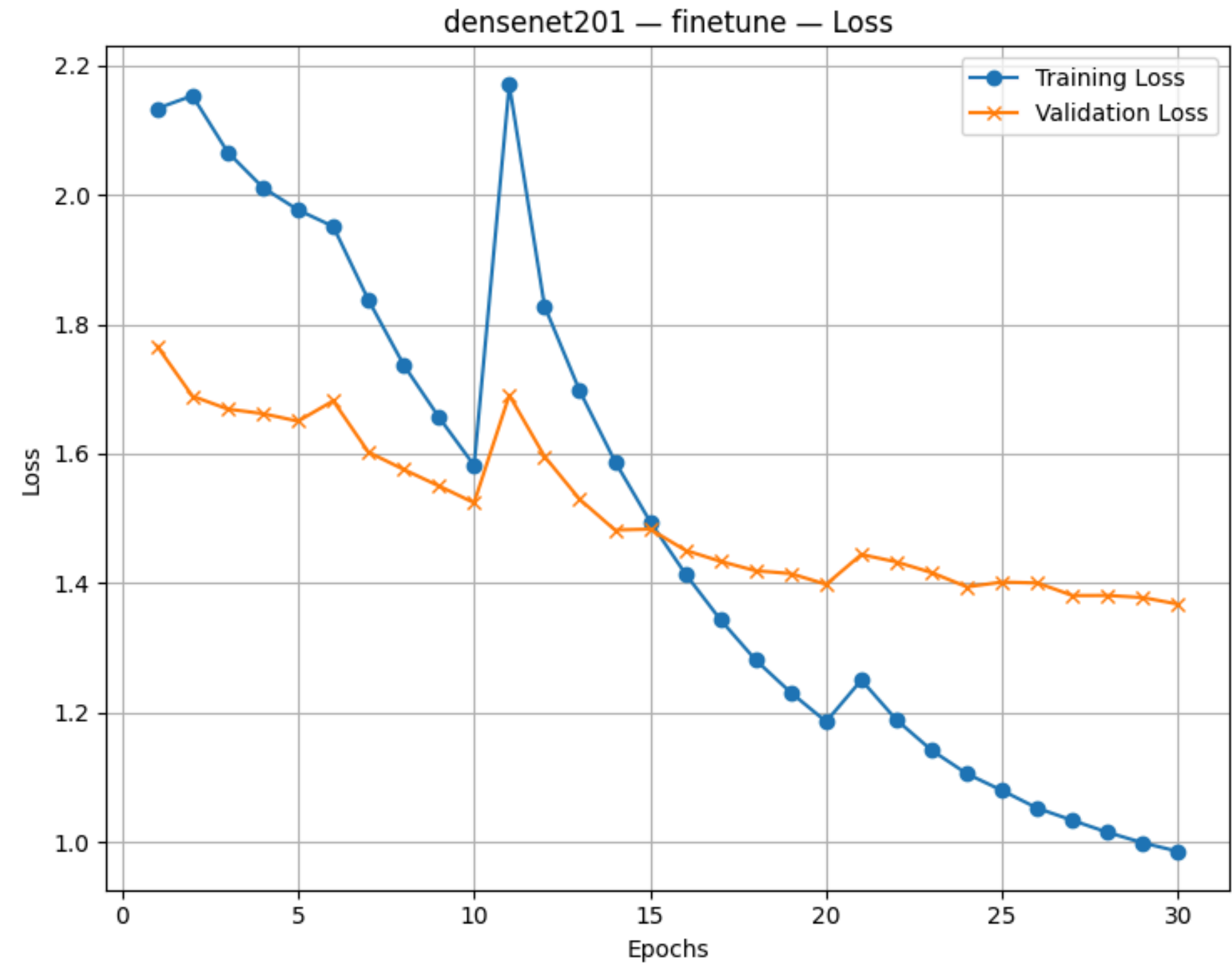
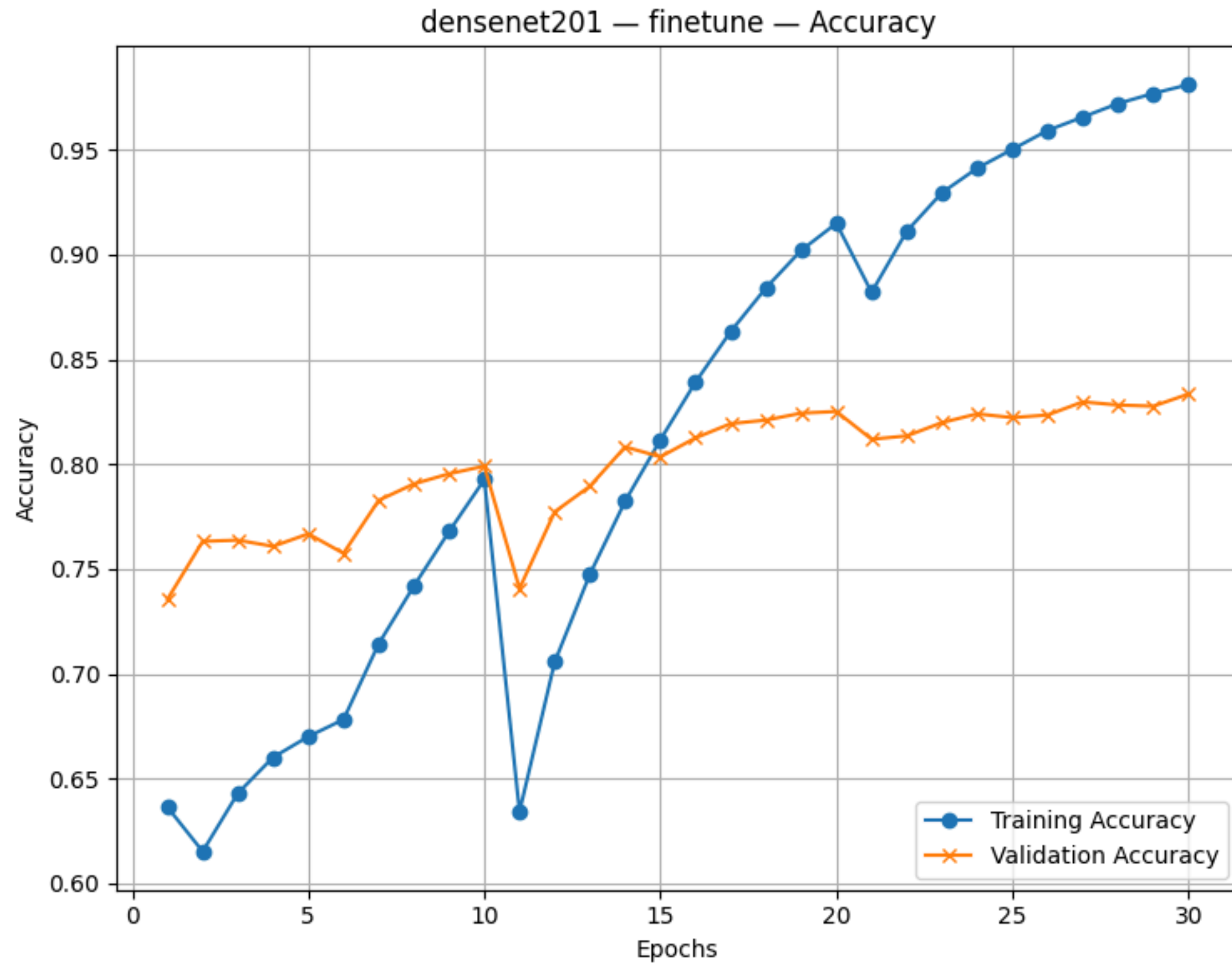
LR_EXP_DECAY = 0.87



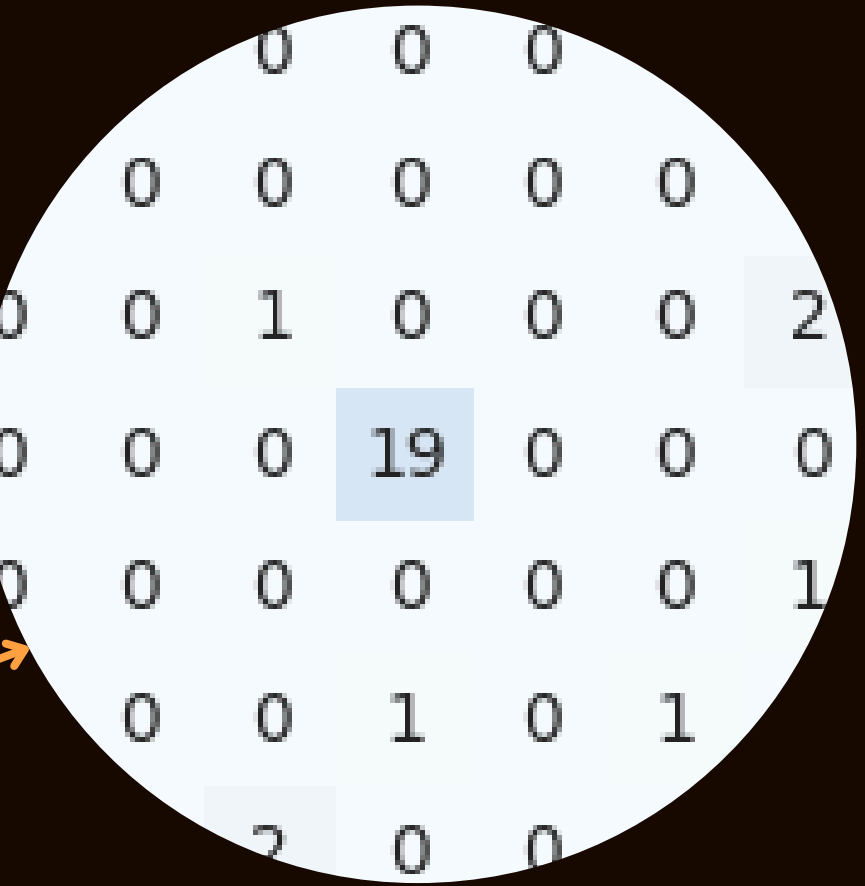
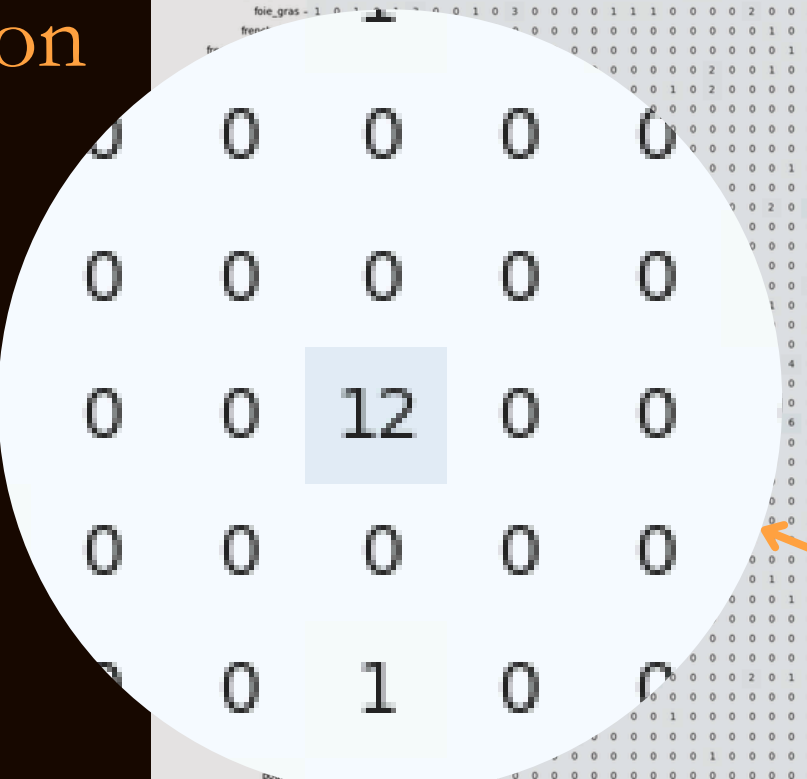
Model Training - Classification Head



Model Training - Fine Tuning



True: Pork Chop
Predicted:
Grilled Salmon



True: Filet Mignon
Predicted: Steak

FINAL EVALUATIONS AND METRICS

- Test Accuracy - 83.24 %
- Test Top 5 Accuracy - 95.87%
- F1-Score - 83.20%
- Precision Score - 83.35%
- Test Recall - 83.24%

SOTA Metrics on Food101

Validation accuracy - 95.17%

